CDLNet: Collaborative Dictionary Learning Network for Remote Sensing Image Scene Classification

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***Abstract*— In recent years, deep-learning-based methods have been extensively utilized in remote sensing image scene classification and have achieved remarkable performance. The wide geographical coverage and resolution differences of scene images result in significant within-class diversity and between-class similarity, hindering the further improvement of classification accuracy. Attention-based methods automatically estimate the importance of local regions by learning weight assignments, which effectively enhance the feature extraction capability of the network. However, methods that solely rely on the network to automatically learn weight assignments may introduce biases in the attention calculations. By analyzing the specific contribution of local features to the key components of global semantics, we propose a collaborative dictionary learning network (CDLNet). CDLNet utilizes the collaborative representation method to decompose global features into a set of key semantic vectors to guide the attention learning process of the network. Specifically, we design a semantic summarization module (SSM) that reconstructs global semantic features by optimizing a low-redundancy dictionary. Next, we propose a global semantic attention module (GSAM) that calculates the contribution of local features to the global feature key information based on their correlation with the reconstructed key semantic set. Finally, an attention transfer loss is introduced to further enhance the attention of low-level feature maps. The experimental results on three publicly available datasets demonstrate that CDLNet can effectively improve within-class diversity and between-class similarity by optimizing the attention learning of the network, thereby achieving great promotion in comparison with state-of-the-art methods. The implementation is publicly available at https://github.com/liuofficial/CDLNet.**

***Index Terms*— Attention, collaborative representation, convolutional neural networks, remote sensing image scene classification, vision transformer (ViT).**

I. Introduction

REMOTE-SENSING images are a significant data source for earth observation and have found extensive applications in various domains in recent years, such as remote sensing image scene classification (RSISC) [1], [2], [3], object detection [4], hyperspectral image classification [5], and image fusion [6], [7]. RSISC is the fundamental task of remote sensing image interpretation that aims to assign unique high-level semantic labels to images based on their ground objects and spatial structure. However, due to the variance and complex spatial distribution of ground objects in scene images, accurately classifying scene images remains a crucial and challenging task [8]. Overall, the challenges of RSISC primarily involve two aspects. On the one hand, there are obvious semantic overlaps between scene images, resulting in high between-class similarity. As shown in Fig. 1(a) and (b), both stadium and playground contain playground objects, while the church and palace buildings exhibit similarities in shape and color. These between-class similarities can easily lead to classification confusion. On the other hand, scene images may exhibit significant within-class diversity, as depicted in Fig. 1(c) and (d).The center category shows noticeable visual

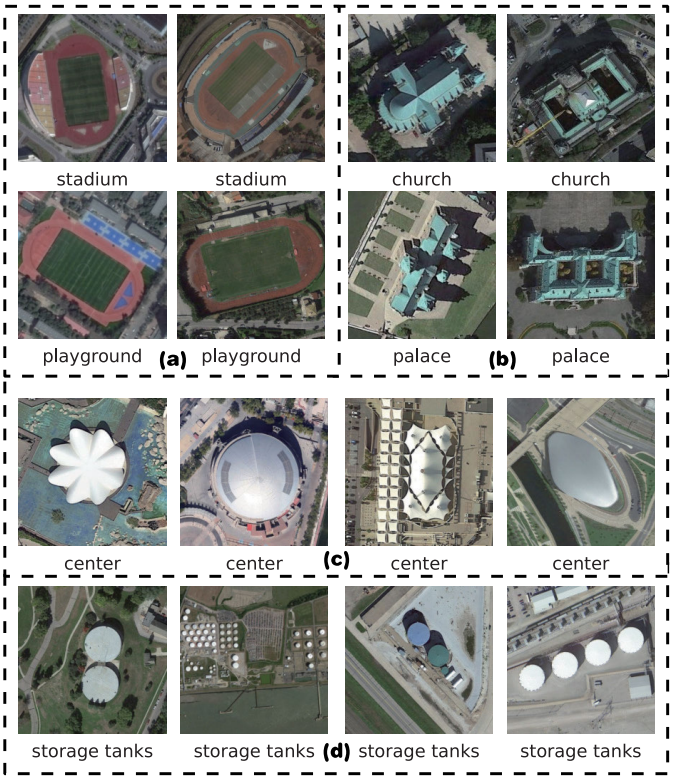


Fig. 1. Main challenges for RSISC include (a) and (b) high between-class similarity and (c) and (d) high within-class diversity.

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differences due to variations in building styles and shapes, while the storage tanks category is influenced by shooting height and object colors, resulting in variations in the size and style of storage tanks. Therefore, extracting more discriminative features from scene images is crucial to address these challenges.

In recent years, deep-learning methods, especially convolutional neural networks (CNNs) [9], [10], [11], have received considerable attention from researchers for their excellent capability to extract high-order features and have found widespread applications in RSISC [2], [3]. However, due to the complex distribution of ground objects in scene images, even CNNs such as VGGNet [9], GoogLeNet [10], ResNet [11], and so on cannot accurately distinguish images with high within-class diversity and between-class similarity. For complex scene images, different local features may have varying contributions to the overall semantic understanding of the image. Previous works [12], [13], [14], [15] often ignore the differences in contributions of different local features when constructing a global representation of images, potentially impeding the identification of key regions in the images. Inspired by the human visual system [16], an attention mechanism [17], [18], [19] has been introduced into computer vision to calculate the importance of different local regions, enabling the model to prioritize critical information in images. Specifically, attention methods estimate the importance of local features by automatically learning weight assignments, allowing the model to minimize attention to irrelevant or noisy parts of the image. However, relying solely on the network to automatically learn weight assignments may introduce biases in the attention calculations, as they do not consider the specific correlation between local features and global semantics. Therefore, it is necessary to learn the weight assignment by analyzing the specific contribution of local features to the global semantic composition.

To address the aforementioned issues, this article proposes a collaborative dictionary learning network (CDLNet), which guides the attention learning process of the network by considering the key semantic components of samples. First, we propose a semantic summarization module (SSM) that summarizes the key semantic information of the sample set by optimizing a shared low-redundancy dictionary. Through the collaborative representation, the dictionary can effectively approximate the global feature of the sample as a set of key semantic vectors. The low-redundancy characteristic helps filter out noncritical semantic information from the sample set, retaining only the most important semantic information. Then, we design a global semantic attention module (GSAM) that estimates the specific contribution of local features in forming global semantic key information by considering their correlation with the reconstructed key semantic set, to obtain more precise attention weights. Furthermore, considering that high-level feature maps have larger receptive fields and can more precisely locate discriminative regions, we introduce an attention transfer loss [20] to transfer attention from high-level to low-level feature maps, thereby further enhancing the discriminative ability of the latter. Finally, we fusion the multilayer feature maps to generate more discriminative and comprehensive representations. In summary, our contributions are as follows.

1) We propose an SSM that summarizes critical semantic information from the sample set by learning a shared low-redundancy dictionary and incorporates the collaborative representation to approximate the global feature of the sample as a set of key semantic vectors.
2) The GSAM is proposed to calculate attention coefficients based on the correlation between local features and the reconstructed key semantic set, thus effectively improving the attention of critical local regions. Furthermore, we introduce an attention transfer loss to further enhance the discriminative ability of low-level feature maps.
3) We propose a CDLNet for RSISC by integrating SSM, GSAM, and attention transfer loss. By enhancing the attention on critical local regions and aggregating multilayer feature maps, CDLNet can obtain a more comprehensive feature representation, thus improving the performance of scene classification.

The structure of this article is as follows. Section II briefly introduces the RSISC method and the collaborative representation related to this article. In Section III, the proposed CDLNet is described in detail. Section IV presents and analyzes the experimental results of CDLNet on three datasets. Finally, the conclusions are summarized in Section V.

II. Related Work

A. RSISC

1) Handcrafted Features Methods: The early scene classification methods mainly rely on handcrafted features methods to extract spectral, color, and texture from images as features for classification [21], [22], [23]. Typical handcrafted feature methods are scale-invariant feature transformation (SIFT) [24] and histogram of oriented gradients (HOG) [25]. These methods often need to combine midlevel descriptors such as visual bag-of-words (BoVW) [1] and vector of locally aggregated descriptors (VLAD) [26] to acquire global information. Handcrafted feature methods struggle to extract rich semantic features of images and require domain knowledge to choose appropriate features. To overcome the limitations of handcrafted features methods, some unsupervised methods have been introduced into scene classification, such as k-means clustering [27], sparse coding [28], and autoencoder [29]. However, these methods have limited discriminative performance, making it difficult to meet the needs of complex scene images.

2) CNN- and ViT-Based Methods: In recent years, deep-learning methods, especially CNNs [11], have emerged as dominant techniques in various domains of natural image analysis due to their remarkable feature extraction capabilities. CNN-based methods have also achieved significant success in RSISC [12], [15], [30], [31], [32], [33]. To reduce the within-class diversity and between-class similarity of scene images, various methods have been explored. Cheng et al. [13] proposed to enhance the discriminative ability of CNNs by

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imposing constraints on the spatial distance of global semantics by metric learning. Wang et al. [34] introduced a weakly supervised key area detection strategy, which addresses the large-scale variance problem in remote sensing scene images by extracting key local features. Li et al. [35] presented a key filter bank method to preserve global information while extracting key local region features. Considering that multiscale information is beneficial for RSISC, He et al. [14] introduced using skip connections and covariance pooling to aggregate multiscale information from feature maps across different levels, thus effectively capturing richer statistical information. In addition, to improve the problem that downsampling in CNNs may lead to loss of texture information, Zhang et al. [36] introduced a trainable Laplacian operator to extract high-frequency features from scene images, which effectively preserved the texture information.

CNNs typically require stacking convolutional layers to increase the receptive field, making it challenging to establish direct relationships between distant local features. The transformer [37] was originally proposed in the field of natural language processing and has achieved significant success. It effectively establishes associations between tokens by mapping input to a query and a set of key–value pairs. The detailed principle of the transformer can be found in [37]. Vision transformer (ViT) [38] introduced the self-attention mechanism [38] into the field of computer vision to solve the problem of insufficient modeling of CNNs and achieved remarkable results. ViT has also been extensively explored in RSISC. Bazi et al. [39] introduced the ViT to scene classification for the first time and demonstrated the potential and feasibility of ViT-based methods in RSISC. Tang et al. [40] proposed an efficient multiscale transformer to extract richer multiscale and contextual information from remote scene images. Lv et al. [41] introduced a progressive aggregation strategy to effectively preserve the local structural information of images while reducing the number of model parameters through iteratively aggregating adjacent tokens. Ma et al. [42] proposed a dual-branch feature learning module based on the transformer, which integrates more effective semantic information by mining homogeneous and heterogeneous information in scene images. Although these methods have made significant advancements in RSISC, significant semantic overlap, and within-class variance still pose certain challenges for classification.

3) Attention-Based Methods: In recent years, inspired by the human visual system, attention mechanisms have been introduced into deep learning to enhance the performance of CNNs in large-scale classification tasks [17], [18], [19], [37], [43]. The attention mechanism adaptively adjusts feature weights based on the importance of the input to discern the significance of each feature. Hu et al. [18] proposed a squeeze-and-excitation (SE) block that adaptively adjusts the channel response by establishing channel-wise associations. Woo et al. [19] proposed a plug-and-play convolutional block attention module (CBAM) that combines spatial and channel attention to learn emphasized or suppressed content and locations, effectively improving the representation ability of CNNs. The attention mechanism has also been widely employed in the field of remote sensing image classification [44], [45], [46], [47], [48], [49], [50], [51]. Chen et al. [44] combined channel attention and spatial attention to propose a local attention module (AM) and embedded it into the residual blocks of ResNet to improve the feature representation ability of the network. Wang et al. [45] designed a recurrent attention structure that utilizes high-level features to generate multiple attention representations, which can effectively extract key local region features. Tang et al. [46] introduced an attention-consistent model that unifies the attention regions of different views, thus effectively improving between-class differences and within-class variations. Tong et al. [47] proposed a channel AM to enhance the discriminative ability of the model and introduced a label smoothing method to improve the issue of between-class similarity in scene images. Although these methods have achieved remarkable results, there is still room for further exploration and refinement in the application of attention mechanisms to RSISC.

B. Collaborative Representation Classification (CRC)

The sparse representation classification (SRC) method [52] has been widely studied as a robust method for face recognition in the early years. The CRC method [53] improves upon SRC by using the ℓ_2-norm, which reduces computational costs while maintaining comparable classification accuracy. Assuming there are M categories, the dictionary set is defined as $\mathbf{D} = [\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^M]$, where $\mathbf{D}^i = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_{N_i}] \in \mathbb{R}^{C \times N_i}$, C is the embedding dimension of atoms, and N_i is the number of training samples in the ith category. In CRC, a test sample $\mathbf{x}$ can be collaboratively represented by the dictionary $\mathbf{D}$ through the following model:

$$\hat{\boldsymbol{s}} = \arg\min_{\mathbf{s}} \frac{1}{2}\|\mathbf{x} - \mathbf{D}\mathbf{s}\|_2^2 + \frac{\lambda}{2}\|\mathbf{s}\|_2^2 \tag{1}$$

where λ is a regularization parameter, $\hat{\boldsymbol{s}} = [\hat{\boldsymbol{s}}^1; \hat{\boldsymbol{s}}^2; \ldots; \hat{\boldsymbol{s}}^M]$ is the sparse coefficient vector with $\hat{\boldsymbol{s}}^i$ corresponding to the ith category. The closed-form analytic solution for the above optimization problem can be formulated as follows:

$$\hat{\boldsymbol{s}} = (\mathbf{D}^T\mathbf{D} + \lambda\mathbf{I})^{-1}\mathbf{D}^T\mathbf{x}. \tag{2}$$

After obtaining $\hat{\boldsymbol{s}}$, the residual between $\mathbf{x}$ and its approximation of each category can be calculated. The label assigned to a sample is determined based on the category with the smallest residual. The classification rule of CRC can be formulated as follows:

$$y = \arg\min_{m=1,2,\ldots,M} \frac{\|\mathbf{x} - \mathbf{D}^m\hat{\boldsymbol{s}}^m\|_2}{\|\hat{\boldsymbol{s}}^m\|_2}. \tag{3}$$

The CRC method is also widely used for RSISC [54], [55]. Mei et al. [54] proposed an efficient fusion strategy to combine multilayer features extracted by CNNs and use CRC for classification. Chen et al. [55] combined the kernel method to further improve the distinguishability of features in high-dimensional space. However, such methods rely heavily on the number of atoms in the dictionary, and the process of collaborative representation does not participate in backpropagation, which limits its further combination with deep methods.

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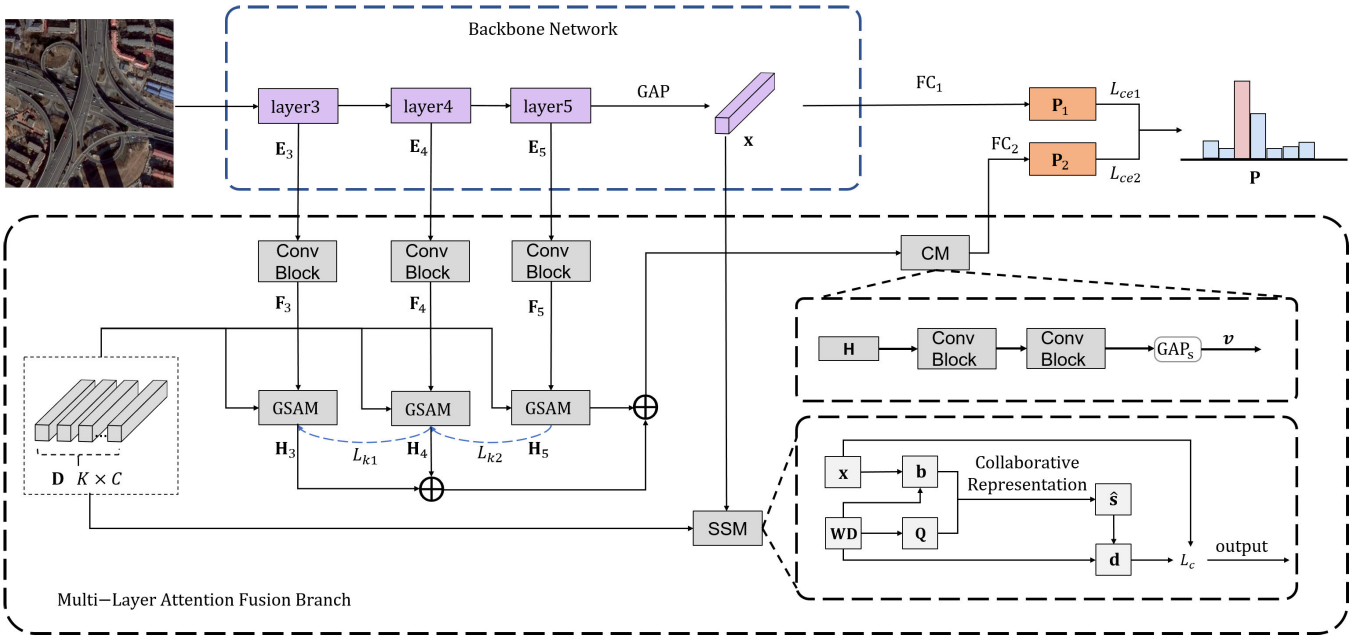


Fig. 2. Framework of the proposed CDLNet. GSAM is a global semantic AM. SSM is a semantic summarization module. Conv Block is a convolutional block. The ⊕ represents addition.

III. Method Description

A. Backbone Network

The proposed CDLNet is shown in Fig. 2, which consists of a backbone network and a multilayer attention fusion branch. We adopt two representative architectures, ResNet-50 and ViT-B, as the backbone network for CDLNet. ViT-B consists of 12 transformer blocks. Since the classification of CDLNet does not depend on the class token, we discard the class token of all layers and select the outputs of the sixth, ninth, and 12th layers as $\mathbf{E}_3$, $\mathbf{E}_4$, and $\mathbf{E}_5$ by default. Then, we reshape them from $N \times C$ to $H \times W \times C$, where N is the number of tokens, $H = W = \sqrt{N}$, and C is the number of channels. For ResNet-50, we select the outputs of the third, fourth, and fifth residual blocks as $\mathbf{E}_3$, $\mathbf{E}_4$, and $\mathbf{E}_5$, respectively.

B. Multilayer Attention Fusion Branch

The multilayer attention fusion branch mainly consists of two modules, SSM and GSAM, and an attention transfer loss. The SSM is employed to learn a shared dictionary that can collaboratively represent each sample in the sample set. The GSAM combines the dictionary learned by SSM to enhance the attention of key regions of multilayer feature maps by computing the specific contribution of local features to global semantics. In addition, we use the attention transfer loss L_k to transfer the attention from high-level feature maps to low-level feature maps, thereby enhancing the representation ability of the latter.

As shown in Fig. 2, the feature maps $\mathbf{E}_3$ and $\mathbf{E}_4$ are first transformed into the consistent spatial and channel dimensions of $\mathbf{E}_5$ by using a convolutional block. It can be formulated as follows:

$$\mathbf{F}_3 = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{n_1,n_1}(\mathbf{E}_3)))$$
$$\mathbf{F}_4 = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{n_2,n_2}(\mathbf{E}_4)))$$
$$\mathbf{F}_5 = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{1,1}(\mathbf{E}_5))) \tag{4}$$

where $\mathrm{Conv}_{n,m}(\cdot)$ represents a convolution operation with a kernel size of $n \times n$ and a stride of m. In CDLNet (ResNet-50), n_1 and n_2 are set to 4 and 2, while in CDLNet (ViT-B), both n_1 and n_2 are set to 1. $\mathrm{BN}(\cdot)$ stands for batch normalization, and $\mathrm{ReLU}(\cdot)$ is a nonlinear activation function. Then, we feed the multilayer features into $\mathrm{GSAM}(\cdot)$ to enhance key region weights. It can be formulated as follows:

$$\mathbf{H}_i = \mathrm{GSAM}(\mathbf{F}_i), \quad i = 3, 4, 5. \tag{5}$$

Finally, we fuse the feature maps $\mathbf{H}_3$, $\mathbf{H}_4$, and $\mathbf{H}_5$ using a simple addition operation and a convolutional module (CM). It can be formulated as follows:

$$\mathbf{H} = \mathbf{H}_3 + \mathbf{H}_4 + \mathbf{H}_5$$
$$\boldsymbol{v} = \mathrm{GAP}_s(\mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3,1}(\mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{1,1}(\mathbf{H})))))))$$
(6)

where $\mathrm{GAP}_s(\cdot)$ represents the global average pooling along the spatial dimension, and $\boldsymbol{v}$ is the output of the multilayer attention fusion branch.

1) Semantic Summarization Module: The SSM aims to learn a dictionary $\mathbf{D} = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_K]$ that can collaboratively represent each sample in the sample set, where K is the number of atoms in the dictionary. First, given a dictionary $\mathbf{D}$ and feature vector $\mathbf{x}$, we get a set of coefficients through collaborative representation. It can be defined as follows:

$$\hat{\boldsymbol{s}} = \arg\min_{\mathbf{s}} \frac{1}{2}\|\mathbf{x} - \mathbf{WDs}\|_2^2 + \frac{\lambda}{2}\|\mathbf{s}\|_2^2 \tag{7}$$

where $\hat{\boldsymbol{s}} = [\hat{s}_1, \hat{s}_2, \ldots, \hat{s}_K]^T$ is the coefficient vector and $\mathbf{W}$ is a learnable linear transformation matrix. According to the least-square method, we can obtain

$$\hat{\boldsymbol{s}} = (\mathbf{Q} + \lambda\mathbf{I})^{-1}\mathbf{b} \tag{8}$$

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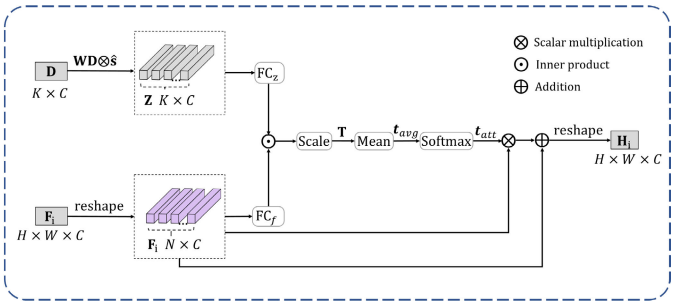


Fig. 3. Detailed structure diagram of GSAM. The transformation matrix $\mathbf{W}$ and the coefficient $\hat{\boldsymbol{s}}$ are derived from the SSM.

where $\mathbf{Q} = (\mathbf{WD})^T(\mathbf{WD})$ and $\mathbf{b} = (\mathbf{WD})^T\mathbf{x}$. λ is set to be 5 by default. Then, we use $\hat{\boldsymbol{s}}$ to combine dictionary atoms to represent $\mathbf{x}$. The reconstructed feature vector can be written as $\mathbf{d} = \mathbf{WD}\hat{\boldsymbol{s}}$. Next, the dictionary $\mathbf{D}$ is optimized by minimizing the distance loss L_c between the reconstructed feature vector $\mathbf{d}$ and $\mathbf{x}$. We normalize $\mathbf{x}$ and $\mathbf{d}$ with the ℓ_2-norm. The loss function L_c can be formulated as follows:

$$L_c = \left\| \frac{\mathbf{x}}{\|\mathbf{x}\|_2} - \frac{\mathbf{d}}{\|\mathbf{d}\|_2} \right\|_2^2. \tag{9}$$

L_c aims to optimize the dictionary $\mathbf{D}$ to better represent the sample feature $\mathbf{x}$. Notably, to stabilize the learning of dictionary $\mathbf{D}$, the loss function L_c does not involve backpropagation on $\mathbf{x}$ and the dictionary $\mathbf{D}$ is initialized randomly.

2) Global Semantic AM: The GSAM enhances key local regions of feature maps by incorporating the dictionary learned by SSM. We combine the parameter matrix $\mathbf{W}$, sparse coefficients $\hat{\boldsymbol{s}}$, and dictionary $\mathbf{D}$ in SSM to compute the reconstructed key semantic set $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_K]$ of feature $\mathbf{x}$. $\mathbf{z}_i = \mathbf{W}\mathbf{d}_i \otimes \hat{s}_i$, where $\otimes$ represents scalar multiplication. As shown in Fig. 3, for the feature map $\mathbf{F}_i \in \mathbb{R}^{H\times W\times C}$, we first reshape it from $H \times W \times C$ to $N \times C$. Then, we apply a fully connected (FC) layer to $\mathbf{F}_i$ and $\mathbf{Z}$ for feature mapping, that is, $\mathbf{F}_i' = \mathrm{FC_f}(\mathbf{F}_i)$ and $\mathbf{Z}' = \mathrm{FC_z}(\mathbf{Z})$. Next, we estimate the contribution of each local feature vector based on the correlation between the local feature vector and the key semantic set $\mathbf{Z}$. Inspired by the transformer [37], we divide each correlation coefficient by $\sqrt{C}$ to ensure a variance of 1. It can be formulated as follows:

$$\begin{aligned} \mathbf{T} &= [\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_N] \\ \mathbf{t}_j &= [t_{j,1}, t_{j,2}, \ldots, t_{j,K}]^T \\ &= \left[\frac{\langle \mathbf{f}'_{i,j}, \mathbf{z}'_1 \rangle}{\sqrt{C}}, \frac{\langle \mathbf{f}'_{i,j}, \mathbf{z}'_2 \rangle}{\sqrt{C}}, \ldots, \frac{\langle \mathbf{f}'_{i,j}, \mathbf{z}'_K \rangle}{\sqrt{C}} \right]^T \end{aligned} \tag{10}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product. The correlation vector $\mathbf{t}_{\mathrm{avg}} \in \mathbb{R}^N$ is obtained by calculating the column mean of $\mathbf{T}$, which indicates the contribution of each local feature vector in constituting the global semantic. It can be formulated as follows:

$$\begin{aligned} \mathbf{t}_{\mathrm{avg}} &= [t_{\mathrm{avg},1}, t_{\mathrm{avg},2}, \ldots, t_{\mathrm{avg},N}] \\ t_{\mathrm{avg},j} &= \frac{1}{K}\sum_{k=1}^{K} t_{j,k}. \end{aligned} \tag{11}$$

Algorithm 1 Training and Inference Process of the CDLNet

Training:
Input: Training set $\boldsymbol{X}_{train}$; A randomly initialized dictionary $\mathbf{D}^0$; Backbone Network $G(\cdot)$; Global Semantic AM GSAM$(\cdot)$; CDLNet parameters θ^0; Training epoch $epoch$;
Output: The optimized CDLNet parameters θ^{epoch}; The optimized dictionary $\mathbf{D}^{epoch}$;
for $i = 1$ to $epoch$ **do**
  $\mathbf{E}_3, \mathbf{E}_4, \mathbf{E}_5, \mathbf{x} \leftarrow G(\boldsymbol{X}_{train})$;
  Get the $\mathbf{F}_3$, $\mathbf{F}_4$, $\mathbf{F}_5$ by (4);
  Get the $\mathbf{H}_3$, $\mathbf{H}_4$, $\mathbf{H}_5$ by (5);
  Get the $\boldsymbol{v}$ by Eq. (6);
  Compute the L_{k1}, L_{k2} using Eq. (12);
  Compute the $\hat{\boldsymbol{s}}$ using Eq. (8);
  Compute the L_c using Eq. (9);
  Compute the L_{ce1}, L_{ce2} using Eq. (14);
  Compute the L_{total} using Eq. (15);
  Optimize parameters θ^i and $\mathbf{D}^i$ with L_{total};
end for
return θ^{epoch} and $\mathbf{D}^{epoch}$;
Inference:
Input: Test set $\boldsymbol{X}_{test}$; Backbone Network $G(\cdot)$; CDLNet parameters θ^{epoch}; dictionary $\mathbf{D}^{epoch}$;
Output: The predicted label $\mathbf{P}$;
  $\mathbf{E_3}, \mathbf{E_4}, \mathbf{E_5}, \mathbf{x} \leftarrow G(\boldsymbol{X}_{train})$;
  Get the $\mathbf{F}_3$, $\mathbf{F}_4$, $\mathbf{F}_5$ by Eq. (4);
  Get the $\mathbf{H}_3$, $\mathbf{H}_4$, $\mathbf{H}_5$ by Eq. (5);
  Get the $\boldsymbol{v}$ by Eq. (6);
  Get the $\mathbf{P}$ by Eq. (13);
return $\mathbf{P}$;

Then, we normalize $\mathbf{t}_{\mathrm{avg}}$ by the Softmax function to obtain the attention coefficient vector $\mathbf{t}_{\mathrm{att}} = \mathrm{Softmax}(\mathbf{t}_{\mathrm{avg}})$. Finally, we multiply the attention coefficient vector with the original feature map and apply residual connection to obtain the feature map $\mathbf{H_i}$ as the output of the GSAM, where $\mathbf{H_i} = \mathbf{F_i} + \mathbf{F_i} \otimes \mathbf{t}_{\mathrm{att}}$.

3) Attention Transfer Loss: High-level feature maps have larger receptive fields, allowing them to focus on discriminative regions more effectively. We introduce an attention loss [20] between adjacent layers, which enables low-level feature maps to gain attention information from high-level feature maps. For feature maps $\mathbf{H}_3$, $\mathbf{H}_4$, and $\mathbf{H}_5$, we use global average pooling to obtain attention maps. The attention transfer loss between $\mathbf{H}_3$, $\mathbf{H}_4$, and $\mathbf{H}_5$ can be formulated as follows:

$$\begin{aligned} \hat{\boldsymbol{H}}_i &= \mathrm{GAP_c}(\mathbf{H}_i), \quad i = 3, 4, 5 \\ L_{k1} &= \left\| \frac{\hat{\boldsymbol{H}}_4}{\|\hat{\boldsymbol{H}}_4\|_2} - \frac{\hat{\boldsymbol{H}}_3}{\|\hat{\boldsymbol{H}}_3\|_2} \right\|_2^2 \\ L_{k2} &= \left\| \frac{\hat{\boldsymbol{H}}_5}{\|\hat{\boldsymbol{H}}_5\|_2} - \frac{\hat{\boldsymbol{H}}_4}{\|\hat{\boldsymbol{H}}_4\|_2} \right\|_2^2 \end{aligned} \tag{12}$$

where $\mathrm{GAP_c}(\cdot)$ represents the global average pooling along the channel dimension. It is worth noting that during backpropagation, the gradients do not flow through $\hat{\boldsymbol{H}}_4$ in L_{k1} and $\hat{\boldsymbol{H}}_5$ in L_{k2}. The attention transfer loss $L_k = L_{k1} + L_{k2}$.

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Fig. 4. Sample images for each category on the UCM dataset.

C. Class Score and Loss Function

We feed the outputs of the backbone and the multilayer attention branch into an FC layer to obtain classification results for M categories. The final classification result $\mathbf{P} \in \mathbb{R}^M$ is obtained by adding the two classification results together. It can be formulated as follows:

$$\mathbf{P} = \mathrm{FC}_1(\mathbf{x}) + \mathrm{FC}_2(\boldsymbol{v}). \tag{13}$$

We supervise the learning of the branch and backbone network with two cross-entropy (CE) losses

$$\begin{aligned} L_{\mathrm{ce}_1} &= -\sum_{k=1}^{M}\left(y_k \cdot \log\left(\frac{\exp(\mathrm{FC}_1(\mathbf{x})_k)}{\sum_{j=1}^{\mathrm{M}} \exp(\mathrm{FC}_1(\mathbf{x})_j)}\right)\right) \\ L_{\mathrm{ce}_2} &= -\sum_{k=1}^{M}\left(y_k \cdot \log\left(\frac{\exp(\mathrm{FC}_2(\boldsymbol{v})_k)}{\sum_{j=1}^{\mathrm{M}} \exp(\mathrm{FC}_2(\boldsymbol{v})_j)}\right)\right) \end{aligned} \tag{14}$$

where y_k represents the ground truth. The total CE loss is $L_{\mathrm{ce}} = L_{\mathrm{ce}1} + L_{\mathrm{ce}2}$. The total loss of CDLNet can be formulated as follows:

$$L_{\mathrm{total}} = L_{\mathrm{ce}} + \alpha L_k + \beta L_c \tag{15}$$

where α and β are used to balance the importance of L_k and L_c. By default, the values of α and β are set to 1 and 5, respectively.

Algorithm 1 provides the pseudocode of the proposed CDLNet. The detailed structure of CDLNet is shown in Table I.

IV. Experiment and Analysis

We evaluate the effectiveness of CDLNet on three publicly available remote sensing scene image datasets using ResNet-50 and ViT-B as backbone networks.

A. Datasets

Three datasets are used to verify the effectiveness of CDLNet, including the UC Merced land-use (UCM) [1] dataset, the aerial image (AID) [2] dataset, and the NWPU-RESISC45 (NWPU) [3] dataset.

1) UC Merced Land-Use Dataset: The UCM dataset, released by the University of California Merced, is one of the most widely used datasets in RSISC. It consists of 21 categories, with each category containing 100 images. The resolution of each image in the dataset is 256 × 256 pixels. Fig. 4 shows some examples of the UCM dataset.

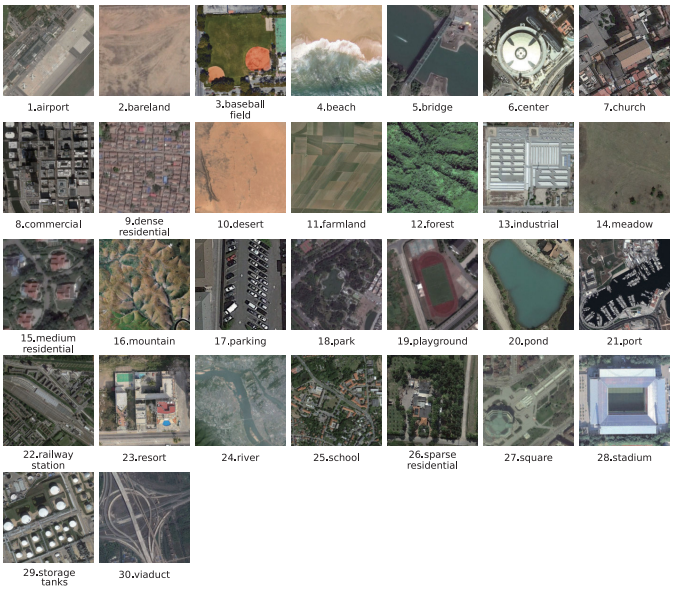


Fig. 5. Sample images for each category on the AID dataset.



Fig. 6. Sample images for each category on the NWPU dataset.

2) AID Dataset: The AID dataset was published by Wuhan University in 2016. It consists of 30 categories, with each category containing 220 to 420 images. The resolution of each image in the dataset is 600 × 600 pixels. Fig. 5 shows some examples of the AID dataset.

3) NWPU-RESISC45 Dataset: The NWPU dataset, constructed by Northwestern Polytechnical University, is one of the largest datasets in RSISC. It consists of 45 different

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TABLE I

DETAILED STRUCTURE OF CDLNET. (FOR BREVITY, ACTIVATION FUNCTIONS AND NORMALIZATION ARE NOT SHOWN. num_cls REPRESENTS THE NUMBER OF CATEGORIES)

Layer name	Output size	CDLNet (ResNet-50)	Output size	CDLNet (ViT-B)
$\mathbf{E}_3$	32×32	7×7 conv, stride $= 2$ 3×3 max pool, stride $= 2$ $\begin{bmatrix} 1 \times 1, & 64 \\ 3 \times 3, & 64 \\ 1 \times 1, & 256 \end{bmatrix} \times 3$ $\begin{bmatrix} 1 \times 1, & 128 \\ 3 \times 3, & 128 \\ 1 \times 1, & 512 \end{bmatrix} \times 4$	14×14	16×16 conv, stride $= 16$ Transformer Block $\times 6$
$\mathbf{E}_4$	16×16	$\begin{bmatrix} 1 \times 1, & 256 \\ 3 \times 3, & 256 \\ 1 \times 1, & 1024 \end{bmatrix} \times 6$	14×14	Transformer Block $\times 3$
$\mathbf{E}_5$	8×8	$\begin{bmatrix} 1 \times 1, & 512 \\ 3 \times 3, & 512 \\ 1 \times 1, & 2048 \end{bmatrix} \times 3$	14×14	Transformer Block $\times 3$
$\mathbf{P}_1$	num_cls	average pool, fc, softmax	num_cls	average pool, fc, softmax
$\mathbf{F}_3$	8×8	4×4 conv, stride $= 4$	14×14	1×1 conv, stride $= 1$
$\mathbf{F}_4$	8×8	2×2 conv, stride $= 2$	14×14	1×1 conv, stride $= 1$
$\mathbf{F}_5$	8×8	1×1 conv, stride $= 1$	14×14	1×1 conv, stride $= 1$
$\mathbf{H}_3$	8×8	GSAM(fc $\times 2$)	14×14	GSAM(fc $\times 2$)
$\mathbf{H}_4$	8×8	GSAM(fc $\times 2$)	14×14	GSAM(fc $\times 2$)
$\mathbf{H}_5$	8×8	GSAM(fc $\times 2$)	14×14	GSAM(fc $\times 2$)
$\mathbf{P}_2$	num_cls	$\begin{bmatrix} 1 \times 1, & 2048 \\ 3 \times 3, & 2048 \end{bmatrix}$ average pool, fc, softmax	num_cls	$\begin{bmatrix} 1 \times 1, & 768 \\ 3 \times 3, & 768 \end{bmatrix}$ average pool, fc, softmax

scene categories, with each category containing 700 images of 256×256 pixels. Fig. 6 shows some examples of the NWPU dataset.

B. Experimental Setup

All experiments are conducted using PyTorch on a workstation equipped with a GeForce RTX 3090 with 24GB memory. The batch size is set to 64. The backbone networks ResNet-50 and ViT-B are initialized with pretrained parameters on ImageNet, while other parts are initialized randomly.

1) ResNet-50: CDLNet (ResNet-50) is optimized using the Adam optimizer [56] for 80 epochs. We use a weight decay of 0.0005 and set the initial learning rate to 0.0001. The learning rate is decreased by a factor of 10 every 30 epochs.

2) ViT-B: For CDLNet (ViT-B), we use the SGD optimizer [57] for 50 epochs with weight decay set to 0.00005 and momentum set to 0.9. The initial learning rate is set to 0.001 and is decreased by a factor of 10 every 20 epochs.

We utilize the same data processing method as ET-GSNet [58]. Data augmentation methods include random flipping and rotation. The input images for ResNet-50 and ViT-B are resized to 256×256 and 224×224, respectively. To ensure result stability, we conducted five repeated experiments with different random seeds and reported the mean overall accuracy (OA) along with the standard deviation (STD). In addition, we also utilize some visualization techniques such as Grad-CAM [59] and t-distributed stochastic neighbor embedding (t-SNE) [60] to analyze the classification results.

C. Comparison With Some SOTA Methods

To evaluate the performance of CDLNet, we compare CDLNet with some SOTA methods, Table II lists the information

TABLE II

DETAILED TYPES AND PUBLICATION YEARS OF THE COMPARED METHODS. (†: CNN-BASED METHODS, ‡: VIT-BASED METHODS, ∗: MLP-BASED METHODS)

Method	Year
GoogLeNet / VGGNet-16 [2] †	TGRS2017
DCNN [13] †	TGRS2018
SCCov [14] †	TNNLS2019
ARCNet [45]	TGRS2019
MIDC-Net_CS [61] †	TIP2020
MG-CAP (Sqrt-E) [15] †	TIP2020
EAM [62] †	GRSL2021
MSA-Network [63] †	JSTARS2021
ACNet [46] †	JSTARS2021
EMTCAL [40] †	TGRS2022
EMSCNet [64] †	TGRS2023
ViT-B [38] ‡	ICLR2021
ET-GSNet [58] ‡	TGRS2022
SCViT [41] ‡	TGRS2022
HHTL [42] ‡	JSTARS2022
Cycle MLP [65] ∗	ICLR2022
CDLNet (ResNet-50) †	Ours
CDLNet (ViT-B) ‡	Ours

of these compared methods. Among them, GoogLeNet [2], VGGNet-16 [2], and ViT-B [38] are baseline methods. DCNN [13] and EMSCNet [64] are methods based on metric learning. SCCov [14], ARCNet [45], MIDC-Net_CS [61], MG-CAP (Sqrt-E) [15], EAM [62], MSA-Network [63], ACNet [46], and EMTCAL [40] are multilayer feature aggregation methods, multigrained methods, and attention-based methods. SCViT [41], HHTL [42], and ET-GSNet [58] are ViT-based methods. Cycle MLP [65] is an MLP-based method.

1) Classification on the UCM Dataset: The comparison results on the UCM dataset are presented in Table III. The CDLNet achieves the highest performance. Specifically, when

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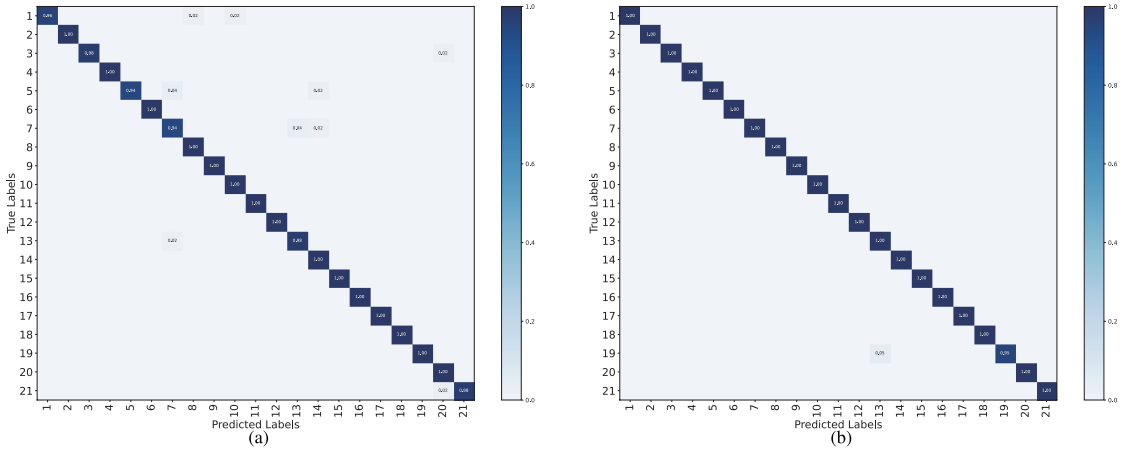


Fig. 7. Confusion matrix of CDLNet (ViT-B) on the UCM dataset when the training ratio is 50% and 80%, respectively. The category numbers are consistent with Fig. 4. (a) 50% training ratio. (b) 80% training ratio.

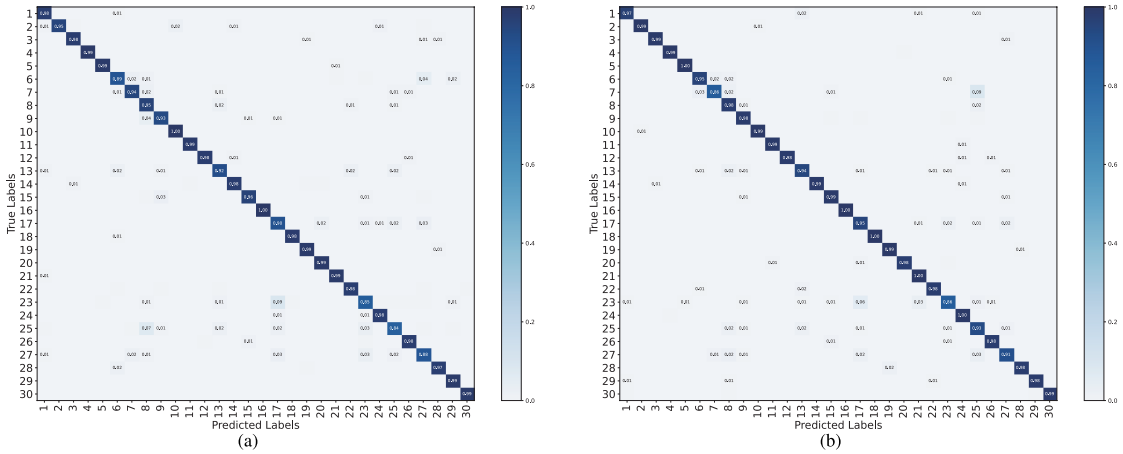


Fig. 8. Confusion matrix of CDLNet (ViT-B) on the AID dataset when the training ratio is 20% and 50%, respectively. The category numbers are consistent with Fig. 5. (a) 20% training ratio. (b) 50% training ratio.

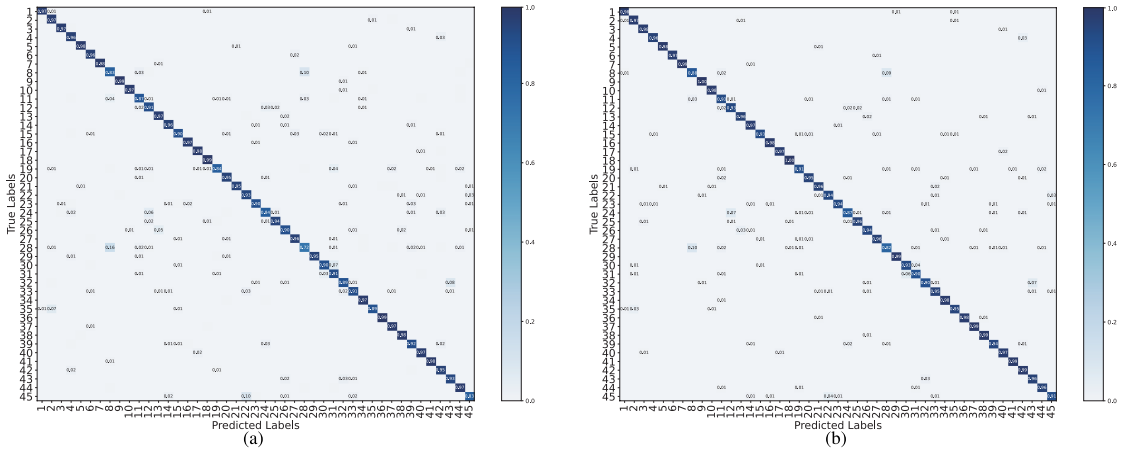


Fig. 9. Confusion matrix of CDLNet (ViT-B) on the NWPU dataset when the training ratio is 10% and 20%, respectively. The category numbers are consistent with Fig. 6. (a) 10% training ratio. (b) 20% training ratio.

trained with 50% data, CDLNet (ViT-B) improves by 0.48% compared to ViT-B, while CDLNet (ResNet-50) improves by 0.73% compared to attention-based methods ACNet [46]. When 80% of the training data is used, existing models already exhibit near-perfect classification accuracy. ELCANet (ViT-B) surpasses all of them by achieving a remarkable classification result of 99.71%. These results demonstrate the effectiveness of the proposed CDLNet on small datasets. In addition, multilayer feature aggregation methods SCCov [14] and EMTCAL [40] have achieved excellent results, which shows that integrating multilayer features can effectively improve classification performance.

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TABLE III

COMPARISON OF OA AND STD ON THE UCM DATASET. THE BEST VALUES ARE MARKED IN BOLD. (†: CNN-BASED METHODS, ‡: VIT-BASED METHODS, ∗: MLP-BASED METHODS)

Method	50% Training Ratio	80% Training Ratio
GoogLeNet [2] †	92.70 ± 0.60	94.31 ± 0.89
VGGNet-16 [2] †	94.14 ± 0.69	95.21 ± 1.20
DCNN [13] †	–	98.93 ± 0.10
SCCov [14] †	–	99.05 ± 0.25
ARCNet [45] †	96.71 ± 0.14	99.12 ± 0.40
MIDC-Net_CS [61]†	95.41 ± 0.40	97.40 ± 0.48
MG-CAP (Sqrt-E) [15]†	–	99.00 ± 0.10
EAM [62] †	98.81 ± 0.26	99.21 ± 0.26
MSA-Network [63] †	97.80 ± 0.33	98.96 ± 0.21
ACNet [46] †	97.99 ± 0.26	99.37 ± 0.15
EMTCAL [40] †	98.67 ± 0.16	99.57 ± 0.28
EMSCNet [64] †	98.70 ± 0.46	99.44 ± 0.16
ViT-B [38] ‡	98.58 ± 0.15	99.28 ± 0.21
ET-GSNet [58] ‡	–	99.29 ± 0.34
SCViT [41] ‡	98.90 ± 0.19	99.57 ± 0.31
HHTL [42] ‡	98.87 ± 0.28	99.48 ± 0.28
Cycle MLP [65] ∗	98.32 ± 0.28	99.05 ± 0.39
CDLNet (ResNet-50) †	98.72 ± 0.17	99.66 ± 0.11
CDLNet (ViT-B) ‡	99.06 ± 0.15	99.71 ± 0.09

TABLE IV

COMPARISON OF OA AND STD ON THE AID DATASET. THE BEST VALUES ARE MARKED IN BOLD. (†: CNN-BASED METHODS, ‡: VIT-BASED METHODS, ∗: MLP-BASED METHODS)

Method	20% Training Ratio	50% Training Ratio
GoogLeNet [2] †	83.44 ± 0.40	86.39 ± 0.55
VGGNet-16 [2] †	86.59 ± 0.29	89.64 ± 0.36
DCNN [13] †	90.82 ± 0.16	96.89 ± 0.10
SCCov [14] †	93.12 ± 0.25	96.10 ± 0.16
ARCNet [45] †	88.75 ± 0.41	93.10 ± 0.55
MIDC-Net_CS [61] †	88.51 ± 0.41	92.95 ± 0.17
MG-CAP (Sqrt-E) [15] †	93.34 ± 0.18	96.12 ± 0.12
EAM [62] †	94.26 ± 0.11	97.06 ± 0.19
MSA-Network [63] †	93.53 ± 0.21	96.01 ± 0.43
ACNet [46] †	93.33 ± 0.29	95.38 ± 0.29
EMTCAL [40] †	94.69 ± 0.14	96.41 ± 0.23
EMSCNet [64] †	95.13 ± 0.10	96.96 ± 0.10
ViT-B [38] ‡	95.21 ± 0.16	96.96 ± 0.22
ET-GSNet [58] ‡	95.58 ± 0.18	96.88 ± 0.19
SCViT [41] ‡	95.56 ± 0.17	96.98 ± 0.16
HHTL [42] ‡	95.62 ± 0.13	96.88 ± 0.21
Cycle MLP [65] ∗	95.31 ± 0.15	96.87 ± 0.17
CDLNet (ResNet-50) †	95.47 ± 0.11	97.19 ± 0.11
CDLNet (ViT-B) ‡	96.07 ± 0.16	97.31 ± 0.12

2) Classification on the AID Dataset: We conducted experiments on the AID dataset, where we randomly selected 20% and 50% of the training data while using the remaining data for testing. The experimental results are shown in Table IV. When using 20% and 50% of the training data, ELCANet (ViT-B) equipped with ViT-B achieved classification accuracies of 96.07% and 97.31%, respectively, which are 0.86% and 0.35% higher than the baseline ViT-B. Furthermore, when ResNet-50 is employed as the backbone network, CDLNet (ResNet-50) consistently achieves higher accuracy at different training ratios compared to the attention-based methods MSA-network [63] and ACNet [46]. This demonstrates the effectiveness of enhancing local feature learning by reconstructing a key semantic set.

3) Classification on the NWPU Dataset: Table V shows the classification results of different methods on NWPU. We can find that CDLNet achieves the best performance under different training ratios. When using 10% training data, CDLNet (ViT-B) achieved 93.43% classification accuracy, 0.71% higher than the second highest SCViT [41]. When using 20% training data, CDLNet (ViT-B) has a classification accuracy of 95.30%, which is 0.80% higher than ET-GSNet [58]. When using ResNet-50 as the backbone, CDLNet (ResNet-50) achieves 92.28% and 94.50% classification results, which is 0.12% and 0.42% higher than EMSCNet [64], respectively. The experimental results show that CDLNet can extract more discriminative features and thus achieve better classification results.

TABLE V

COMPARISON OF OA AND STD ON THE NWPU DATASET. THE BEST VALUES ARE MARKED IN BOLD. (†: CNN-BASED METHODS, ‡: VIT-BASED METHODS, ∗: MLP-BASED METHODS)

Method	10% Training Ratio	20% Training Ratio
GoogLeNet [3] †	76.19 ± 0.38	78.48 ± 0.26
VGGNet-16 [3] †	76.47 ± 0.18	79.79 ± 0.15
DCNN [13] †	89.22 ± 0.50	91.89 ± 0.22
SCCov [14] †	89.30 ± 0.35	92.10 ± 0.25
ARCNet [45] †	91.69 ± 0.39	93.34 ± 0.88
MIDC-Net_CS [61] †	86.12 ± 0.29	87.99 ± 0.18
MG-CAP (Sqrt-E) [15] †	90.83 ± 0.12	92.95 ± 0.13
EAM [62] †	91.91 ± 0.22	94.29 ± 0.09
MSA-Network [63] †	90.38 ± 0.17	93.52 ± 0.21
ACNet [46] †	91.09 ± 0.13	92.42 ± 0.16
EMTCAL [40] †	91.63 ± 0.19	93.65 ± 0.12
EMSCNet [64] †	92.16 ± 0.07	94.08 ± 0.20
ViT-B [38] ‡	92.45 ± 0.12	94.74 ± 0.12
ET-GSNet [58] ‡	92.72 ± 0.28	94.50 ± 0.18
SCViT [41] ‡	92.72 ± 0.04	94.66 ± 0.10
HHTL [42] ‡	92.07 ± 0.44	94.21 ± 0.09
Cycle MLP [65] ∗	91.84 ± 0.17	94.27 ± 0.11
CDLNet (ResNet-50) †	92.28 ± 0.25	94.50 ± 0.19
CDLNet (ViT-B) ‡	93.43 ± 0.18	95.30 ± 0.03

D. Confusion Matrix Analysis

To visually show the degree of confusion, we plot the confusion matrix of CDLNet (ViT-B) on the three datasets in Figs. 7–9, where the row represents the predicted labels and the column represents the true labels. Fig. 7 shows the confusion matrices trained with 50% and 80% data on the UCM dataset. As shown in Fig. 7(a), when using 50% of the training data, all category predictions are above 90% accurate. We also achieved 94% and 98% classification accuracy for the two confusing categories dense residential (7) and medium residential (13), respectively. When using 80% of the training data, as shown in Fig. 7(b), CDLNet (ViT-B) mispredicted only one image of sparse residential (19) as medium residential (13), showing that our method can achieve stable improvements on small dataset.

Fig. 8 shows the confusion matrices trained with 20% and 50% data on the AID dataset. As shown in Fig. 8(a), when using 20% of the training data, we observed that 26 categories achieved a remarkable 90% classification accuracy. Moreover, as shown in Fig. 8(b), when we use 50% of the training data, we still achieved good classification results for challenging categories with higher within-class variance, such as church (7), school (25), and storage tanks (29). This indicates that CDLNet (ViT-B) effectively mitigates the issue

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TABLE VI
COMPARISON RESULTS OF CDLNET (RESNET-50) AND ATTENTION-BASED NETWORKS ON THE AID AND NWPU DATASETS

Method	AID (20%)	AID (50%)	NWPU (10%)	NWPU (20%)
CBAMNet [19]	95.11	96.88	91.83	94.17
ECANet [43]	95.09	96.72	91.79	94.25
SENet [18]	95.16	96.91	91.97	94.20
CDLNet (ResNet-50)	95.47	97.19	92.28	94.50

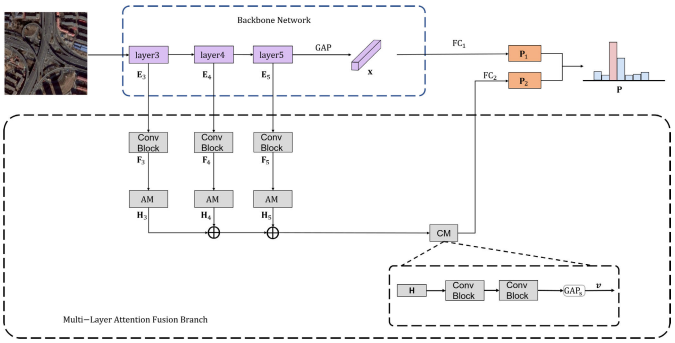


Fig. 10. Architecture of attention-based methods. The AM can use CBAM [19], SE [18], SA [37], or ECA [43], and the other parts are consistent with Fig. 2.

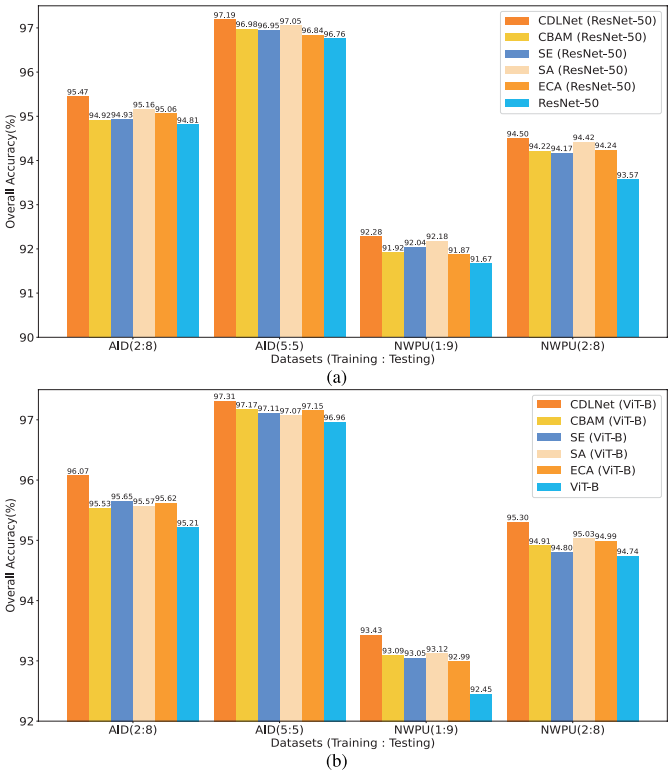


Fig. 11. Comparison results of CDLNet and attention methods on the AID and NWPU datasets. (a) ResNet-50. (b) ViT-B.

of within-class diversity in scene images, leading to improved classification performance.

The confusion matrix of the NWPU dataset is presented in Fig. 9. As shown in Fig. 9(a), when using 10% of the training data, CDLNet (ViT-B) achieves a classification accuracy exceeding 90% in 37 of the 45 categories. When using 20% of the training data, CDLNet (ViT-B) only misclassifies 0.04% of wetland (45) as class lake (22) and 0.07% of rectangular farmland (32) as terrace (43), which shows that CDLNet (ViT-B) can effectively improve the between-class similarity.

E. Comparison of Different Attention Methods

To verify the effectiveness of the proposed method, we compared CDLNet with classical attention methods. First, we employ ResNet-50 as the backbone to compare CDLNet with attention networks CBAMNet [19], ECANet [43], and SENet [18]. The comparison methods follow the default parameter settings in the original paper and use parameters pretrained on ImageNet for network initialization. The experimental results are presented in Table VI. When employing 20% of the training data on the AID dataset, CDLNet (ResNet-50) achieves a classification accuracy that is 0.31% higher than SENet [18]. When using 20% of the training data on the NWPU dataset, CDLNet (ResNet-50) attains a classification accuracy of 94.50%, surpassing ECANet [43] by 0.25%. These results demonstrate the effectiveness of CDLNet in improving scene classification performance. In addition, as shown in Fig. 10, we adopt a multilayer structure consistent with CDLNet, where the AM can use CBAM [19], SE [18], SA [37], or ECA [43]. All experiments are conducted on the AID and NWPU datasets using ViT-B and ResNet-50 as the backbone network. The experimental results are presented in Fig. 11. As shown in Fig. 11(a), ResNet-50 equipped with different AMs has significantly improved compared to the baseline ResNet-50. Our CDLNet (ResNet-50) achieves the best performance. It is worth noting that ResNet-50 equipped with a self-AM achieved the second-best result, indicating that establishing direct dependencies between image patches is beneficial for exploring the context information of scene images and the spatial distribution of objects, leading to improved performance. When equipped with ViT-B, as shown in Fig. 11(b), CDLNet (ViT-B) still achieves advanced performance compared to other methods, which demonstrates the effectiveness of the dictionary learning-based attention method in improving network attention learning.

F. Small Sample Size Experiment

We conducted small sample size experiments on the UCM dataset and compared it with dictionary-based learning methods GLFSRC [54], FFKCRC [55], and attention-based methods CBAMNet [19], ECANet [43], and SENet [18] to evaluate the performance of CDLNet under conditions of limited training data. The experimental results are shown in Table VII. Compared with the dictionary-based learning methods GLFSRC and FFKCRC that do not involve the backpropagation

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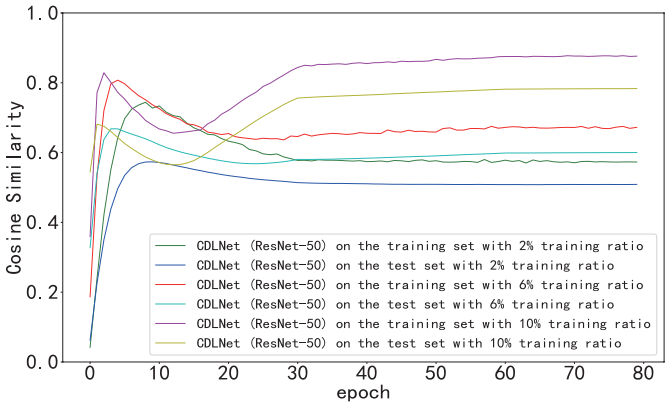


Fig. 12. Cosine similarity (as a function of epoch) between features **x** and reconstructed features **d** is calculated for the UCM dataset using 2%, 6%, and 10% of the training data, respectively.

TABLE VII
SMALL SAMPLE SIZE EXPERIMENT ON THE UCM DATASET. THE BEST VALUES ARE MARKED IN BOLD

Method	(2%)	(4%)	(6%)	(8%)	(10%)
GLFSRC [54]	61.61	73.61	77.71	83.59	87.67
FFKCRC [55]	62.59	76.49	80.45	83.75	87.65
CBAMNet [19]	67.05	82.02	86.12	87.08	89.75
ECANet [43]	66.47	81.81	85.98	86.83	89.59
SENet [18]	66.78	82.37	86.03	87.13	89.91
CDLNet (ResNet50)	65.16	81.25	85.76	87.16	90.26

process, CDLNet can achieve better classification results. The performance of CDLNet depends on the accurate representation of feature **x** by dictionary **D** in the SSM. As shown in Fig. 12, we record the cosine similarity of features **x** and **d** during network training under different training ratios. When the training data is less than 6%, the dictionary **D** cannot adequately represent the feature **x**, causing CDLNet to be unable to effectively perform its performance. When we use 10% of the training data, the dictionary **D** can effectively approximate **x**, thus achieving better results.

G. Ablation Study

1) Effectiveness of SSM: The purpose of the SSM is to learn a shared dictionary **D** that can represent sample feature vector **x**. Its effectiveness depends on whether the feature vector **d** reconstructed through the dictionary can approximately represent **x**. We first verify whether SSM can effectively optimize dictionary **D**. As shown in Fig. 13, we recorded the cosine similarity between the features **x** generated by the backbone network and the features **d** generated by the dictionary **D** during the training and testing process of CDLNet. It can be observed that the average cosine similarity between features **x** and **d** exhibits a continuously increasing trend on the three datasets with iteration. As shown in Fig. 14, we record the curves of loss function L_c during network training. As training progresses, the loss L_c steadily decreases after some initial fluctuations and eventually stabilizes. These results demonstrate that SSM can effectively optimize the dictionary **D** under the supervision of the loss function L_c.

The layer FC_1 is an FC layer employed for the classification of feature **x**. It is optimized by the CE loss between the classification result of feature **x** and the ground truth. To further validate the effectiveness of SSM, we employ FC_1 to classify the features **d**, aiming to assess whether CDLNet can recognize the semantic information of the reconstructed features **d**. As shown in Fig. 15, as the training progresses, FC_1 can effectively classify features **d**, which also shows that features **d** can effectively represent the key semantics of features **x**.

TABLE VIII
EFFECT OF λ ON THE AID AND NWPU DATASETS

λ	AID (20%)	AID (50%)	NWPU (10%)	NWPU (20%)
1	95.25	97.15	92.11	94.47
3	95.34	97.29	92.31	94.56
5	95.47	97.19	92.28	94.50
10	95.43	97.17	92.37	94.63

TABLE IX
ABLATION RESULTS OF THE DIFFERENT COMPONENTS IN CDLNET ON THE AID AND NWPU DATASETS. BRACKETS INDICATE IMPROVEMENTS COMPARED TO THE BASELINE NETWORK. (THE ✓ INDICATES THAT THE MODULE IS RETAINED)

Backbone	GSAM	L_k	AID (20%)	NWPU (10%)
ViT-B	✓	✓	96.07(↑ 0.86)	93.43(↑ 0.98)
	✓		95.85(↑ 0.64)	93.31(↑ 0.86)
		✓	95.53(↑ 0.32)	92.93(↑ 0.48)
			95.21	92.45
ResNet-50	✓	✓	95.47(↑ 0.66)	92.28(↑ 0.61)
	✓		95.21(↑ 0.40)	92.05(↑ 0.38)
		✓	95.18(↑ 0.37)	91.88(↑ 0.21)
			94.81	91.67

2) Regularization Parameter λ: λ is a regularization parameter of SSM, which can stabilize the least-squares solution and introduce a certain degree of sparsity to the solution [53]. We evaluate its impact on CDLNet (ResNet-50) through experiments. As shown in Table VIII, altering the λ value does not lead to significant fluctuations in the classification results, which shows that SSM has high robustness to λ.

3) Effect of Dictionary Size: We analyze the impact of the number of learnable atoms K in the dictionary on the experimental results. As shown in Fig. 16, as the parameter K increases, the overall trend of the accuracy rate increases first and then decreases. This is because a smaller K may not be able to effectively reconstruct the features **x** of the sample set, while a larger value of K may not be conducive to learning a low-redundancy dictionary. By default, we set the value of K to be the same as the number of categories.

4) Effectiveness of Different Modules: To analyze the contribution of different components in CDLNet, we conduct ablation experiments on the AID and NWPU datasets for GSAM and attention transfer loss. The ablation results are shown in Table IX. When we remove the attention transfer loss, the OA of CDLNet (ViT-B) drops by 0.22% and 0.12% on the AID and NWPU datasets, respectively. Moreover, when the GSAM is removed, the OA of CDLNet (ViT-B) drops by 0.54% and 0.50% on the AID and NWPU datasets, respectively. Experimental results show the effectiveness of GSAM and attention transfer loss, and GSAM has a greater impact on the results than attention transfer loss. Furthermore, similar results are obtained when ResNet-50 is employed as

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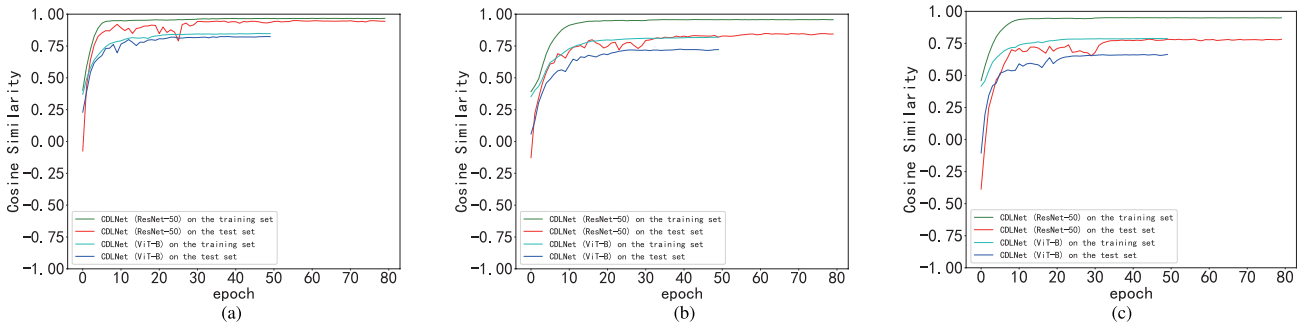


Fig. 13. Cosine similarity (as a function of epoch) between features $\mathbf{x}$ and reconstructed features $\mathbf{d}$ is calculated for the UCM, AID, and NWPU datasets using 80%, 20%, and 10% of the training data, respectively. (a) Cosine similarity (UCM). (b) Cosine similarity (AID). (c) Cosine similarity (NWPU).

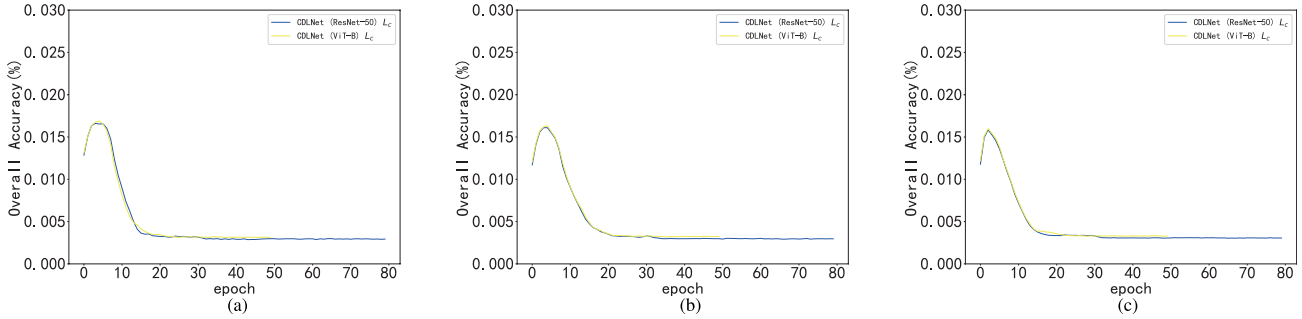


Fig. 14. Curves of the loss function L_c on the UCM, AID, and NWPU datasets with 80%, 20%, and 10% of the training data, respectively. (a) L_c (UCM). (b) L_c (AID). (c). L_c (NWPU).

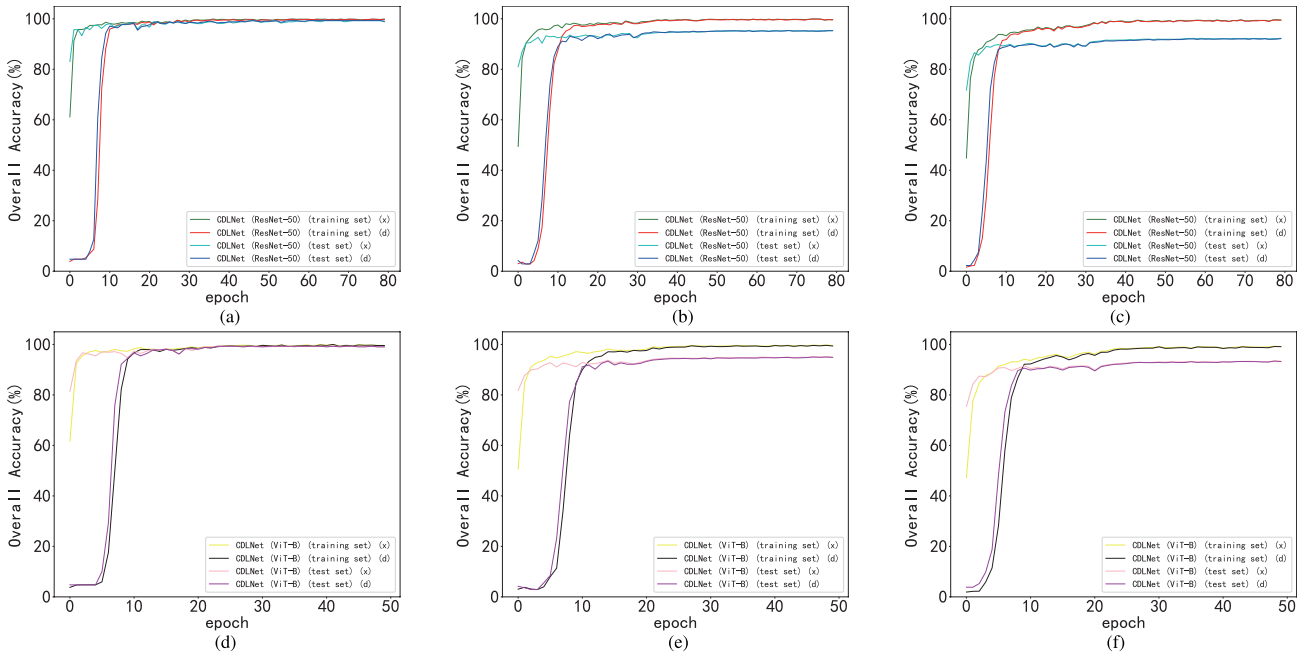


Fig. 15. OA curves for classifying features $\mathbf{d}$ and $\mathbf{x}$ using FC_1 on the UCM, AID, and NWPU datasets with 80%, 20%, and 10% of the training data, respectively. (a) CDLNet (ResNet-50) (UCM). (b) CDLNet (ResNet-50) (AID). (c) CDLNet (ResNet-50) (NWPU). (d) CDLNet (ViT-B) (UCM). (e) CDLNet (ViT-B) (AID). (f) CDLNet (ViT-B) (NWPU).

the backbone, indicating the effectiveness of both GSAM and attention transfer loss for different network architectures.

5) Effect of GSAM Embedding Position: Drawing from insights gained from prior research [14], [31], [40], we use the outputs of the last three blocks of ResNet-50 as $\mathbf{E}_3$, $\mathbf{E}_4$, and $\mathbf{E}_5$. The selection of $\mathbf{E}_3$, $\mathbf{E}_4$, and $\mathbf{E}_5$ for ViT-B is done randomly. To elucidate the effect of GSAM embedding position on CDLNet, we employ CDLNet (ResNet-50) and CDLNet (ViT-B) to assess the variations in experimental results arising from differences in the number and depth of blocks employed, respectively. We first verify the effect of the number of blocks employed on the experimental results of CDLNet (ResNet-50). Scheme 1 embeds a branch after the 2th residual block of the original CDLNet (ResNet-50), and the structure is consistent

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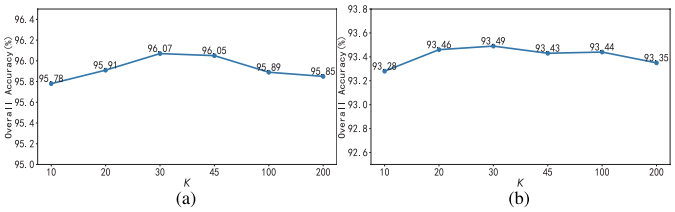


Fig. 16. Effect of dictionary size K for CDLNet (ViT-B) on the AID and NWPU datasets, where the training ratios are 20% and 10%, respectively. (a) K on the AID dataset. (b) K on the NWPU dataset.

TABLE X

EFFECT IN EXPERIMENTAL RESULTS ON CDLNET ARISING FROM DIFFERENCES IN THE NUMBER OF BLOCKS EMPLOYED. (THE NUMBERS IN BRACKETS INDICATE THE LAYERS IN WHICH THE GSAM IS EMBEDDED)

Scheme	AID (20%)	AID (50%)	NWPU (10%)	NWPU (20%)
1 (2,3,4,5)	95.53	97.25	92.31	94.53
2 (3,4,5)	95.47	97.19	92.28	94.50
3 (4,5)	95.43	97.17	92.21	94.43

TABLE XI

EFFECT IN EXPERIMENTAL RESULTS ON CDLNET ARISING FROM DIFFERENCES IN THE DEPTH OF BLOCKS EMPLOYED. ((i, j, k) INDICATES USING THE iTH, jTH, AND kTH LAYERS OF VIT-B AS $\mathbf{E}_3$, $\mathbf{E}_4$, AND $\mathbf{E}_5$, RESPECTIVELY)

Scheme	AID (20%)	AID (50%)	NWPU (10%)	NWPU (20%)
4 (6,9,12)	96.07	97.31	93.43	95.30
5 (8,10,12)	96.11	97.33	93.51	95.43
6 (10,11,12)	96.16	97.39	93.59	95.47

with the branch where $\mathbf{F}_3$ is located, that is, GSAM is embedded after the second, third, fourth, and fifth residual blocks. Scheme 2 corresponds to the original CDLNet (ResNet-50), that is, GSAM is embedded after the 3th, 4th, and 5th residual blocks. Scheme 3 removes the branch where $\mathbf{F}_3$ is located in the original CDLNet (ResNet-50), that is, GSAM is embedded after the fourth and fifth residual blocks. The experimental results of Schemes 1–3 correspond to lines 1–3 in Table X, respectively. When additionally embedding a branch after the 2th residual block, CDLNet (ResNet-50) can achieve better results. When the branch located after the third residual block is removed, the accuracy of CDLNet (ResNet-50) decreases slightly. These results demonstrate that embedding GSAM on more residual blocks can improve the OA of CDLNet (ResNet-50). We subsequently verify the effect of the depth of blocks employed on the experimental results of CDLNet (ViT-B). Scheme 4 is the original CDLNet (ViT-B). Scheme 5 uses the outputs of the eighth, tenth, and 12th of ViT-B as $\mathbf{E}_3$, $\mathbf{E}_4$, and $\mathbf{E}_5$, respectively. Scheme 6 uses the outputs of the tenth, 11th, and 12th of ViT-B as $\mathbf{E}_3$, $\mathbf{E}_4$, and $\mathbf{E}_5$, respectively. The experimental results of Schemes 4–6 correspond to lines 1–3 in Table XI, respectively. Compared with the original CDLNet (ViT-B), higher classification accuracy can be achieved using Schemes 5 and 6. These results demonstrate that using deeper blocks, CDLNet (ViT-B) can achieve better results.

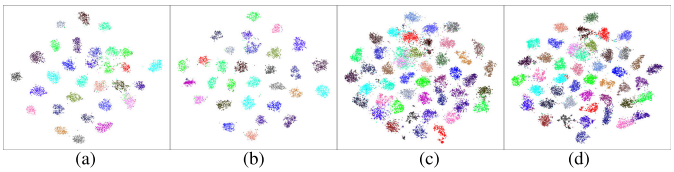


Fig. 17. t-SNE results on the AID and NWPU datasets, where the training ratios are 20% and 10%, respectively. (a) ViT-AID. (b) CDLNet-AID. (c) ViT-NWPU. (d) CDLNet-NWPU.

H. Visualization Experiment

The t-SNE is a dimensionality reduction technique commonly used to visualize high-dimensional data in a lower-dimensional space. We present the visualization results of ViT-B and CDLNet (ViT-B) on the AID and NWPU datasets by t-SNE. As shown in Fig. 17, CDLNet (ViT-B) exhibits more compact cluster centers and a larger distance between clusters compared to ViT-B. These results demonstrate that our proposed method effectively reduces within-class diversity and between-class similarity.

The Grad-CAM [59] is widely used in network visualization, which can intuitively reflect the ability of the network to locate discriminative regions. We utilized the Grad-CAM method to generate the $\mathbf{E}_5$ energy maps of ViT-B and CDLNet (ViT-B). As shown in Fig. 18, for the storage tanks, church, and center categories, CDLNet (ViT-B) can more accurately locate salient objects than ViT-B. Both the playground and the stadium contain playground buildings and CDLNet (ViT-B) distinguishes these two categories through the information around the playground. For the sparse residential and forest categories, CDLNet (ViT-B) utilizes information such as vegetation density and buildings to discriminate between these two categories. The lake and wetland are easily confused, and CDLNet (ViT-B) distinguishes them through lake and vegetation information. These results indicate that CDLNet (ViT-B) can extract more precise and discriminative information compared to VIT-B, thereby reducing within-class diversity and between-class similarity. Furthermore, as illustrated in Fig. 19, we generate some layer-wise energy maps for CDLNet (ViT-B). In the case of the stadium category, the energy map $\mathbf{E}_5$ is mainly to observe the stadium building. The energy map $\mathbf{H}_5$ not only emphasizes the stadium building, but also further integrates the contextual information from the surrounding buildings. By combining feature maps $\mathbf{E}_5$ and $\mathbf{H}_5$, CDLNet can generate more comprehensive semantic features. For the storage tanks category, the energy map $\mathbf{H}_5$ can recognize more storage tank objects than $\mathbf{E}_5$, which shows that our proposed method can effectively enhance the recognition ability of discriminative features. For the baseball field and center categories, the energy map $\mathbf{H}_4$ can locate key regions earlier than $\mathbf{E}_4$, which shows that the multilayer attention fusion branch can effectively enhance the perceptual ability of low-level feature maps for discriminative regions.

I. Number of Parameters and Computation Complexity

Table XII shows the computational complexity and parameter comparison results of CDLNet and some attention methods on the AID and NWPU datasets. We utilize ViT-B as the

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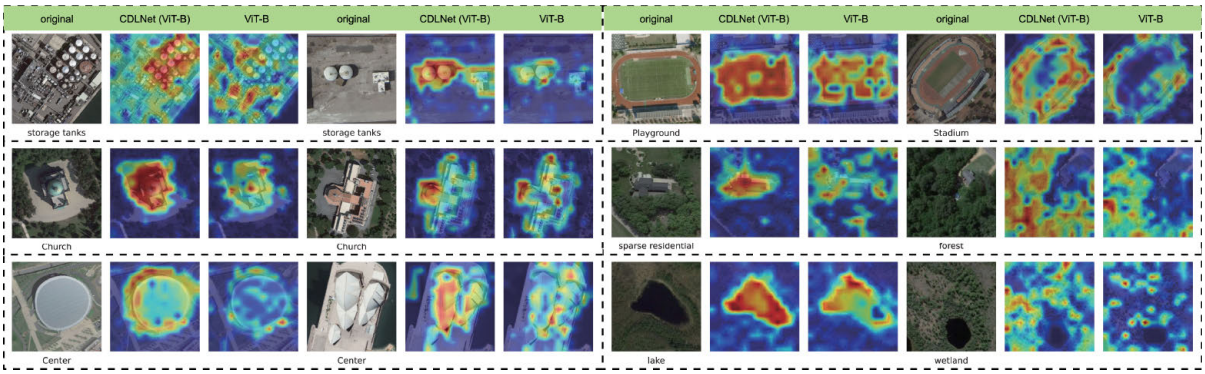


Fig. 18. Some samples of energy maps based on the fine-tuned ViT-B and the proposed CDLNet (ViT-B).

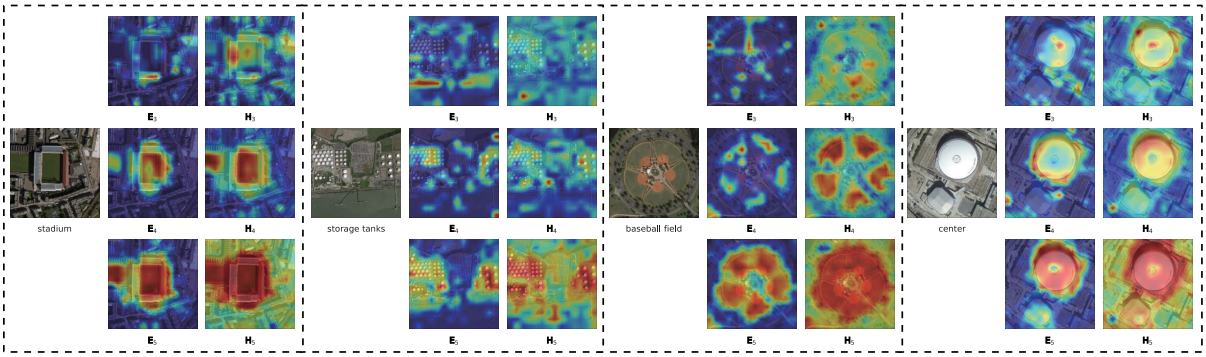


Fig. 19. Layer-wise energy map based on the proposed CDLNet (ViT-B).

TABLE XII
NUMBER OF PARAMETERS AND COMPUTATION COMPLEXITY FOR ONE BATCH ON THE AID AND NWPU DATASETS

Method	Parameters	Train time	Test time	AID (20%)	NWPU (10%)
ViT-B [38]	85.63M	0.469s	0.165s	95.21	92.45
CBAM (ViT-B) [19]	96.42M	0.512s	0.174s	95.53	93.09
SE (ViT-B) [18]	96.64M	0.504s	0.172s	95.65	93.05
ECA (ViT-B) [43]	96.34M	0.506s	0.177s	95.62	92.99
SA (ViT-B) [37]	103.43M	0.624s	0.213s	95.57	93.12
CDLNet (ViT-B)	103.02M	0.568s	0.199s	96.07	93.43

backbone network to measure the training and test time of a batch of data within an epoch. Compared with the baseline model, the classification performance of the methods using different AMs is improved, while the number of parameters and computation time are also increased. CDLNet has a slight increase in computational complexity and parameters, but it achieves more significant performance improvements.

V. CONCLUSION

In this article, we design a CDLNet for RSISC. CDLNet guides the network to focus on salient regions of the image through the dictionary-learned attention method and integrates multilayer feature maps to extract more discriminative representations. Specifically, we propose an SSM to reconstruct sample features by learning a shared dictionary via collaborative representation. Then, GSAM is subsequently proposed, which is combined with SAM to calculate the specific contribution of different local regions to the global feature key information. Furthermore, we introduce an attention transfer loss to further enhance the representation ability of low-level feature maps. We evaluate the effectiveness of CDLNet equipped with ResNet-50 and ViT-B on three datasets. The experimental results show that the dictionary learning-based attention method is more effective in extracting key features compared to traditional attention methods. Compared with SOTA methods, CDLNet can achieve better results in RSISC. In future work, we intend to further delve into dictionary learning methods to improve the performance of RSISC.

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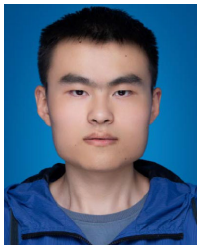
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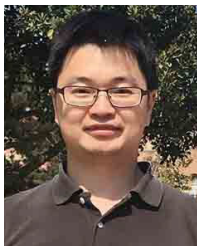
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